CONFIDENTIAL – INTERNAL USE

WestBrom External Backup GCP

Infrastructure Architecture Design & Technical Diagrams

Converge Banking (CB) · Cross-Cloud Off-Site Backup Solution

JIRA REFERENCE	DOCUMENT DATE
CS-14029	April 2026

GCP REGION	NON-PROD TARGET
europe-west2 (London)	30 April 2026

MANAGED BY
Terraform IaC

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1. Solution Overview

This solution deploys the **GCP-side infrastructure** for the WestBrom (Converge Banking) cross-cloud off-site backup strategy. Aurora PostgreSQL backups and Cognito backups are transferred nightly from **AWS S3** to **GCP Cloud Storage** via the Storage Transfer Service, providing cloud-level isolation in the event of a widespread AWS regional outage.

☁️

SOURCE CLOUD

Amazon AWS

Aurora PostgreSQL + Cognito

🎯

DESTINATION

Google Cloud

europe-west2 (London)

⏱️

SCHEDULE

04:00 UTC

Daily (every 86400s)

🔒

RETENTION

30 Days

Delete at 90 days

🔄

RTO / RPO

7d / 24h

Widespread AWS outage

🏗️

DEPLOYMENT

Terraform IaC

Azure DevOps Pipeline

GCP Services Used

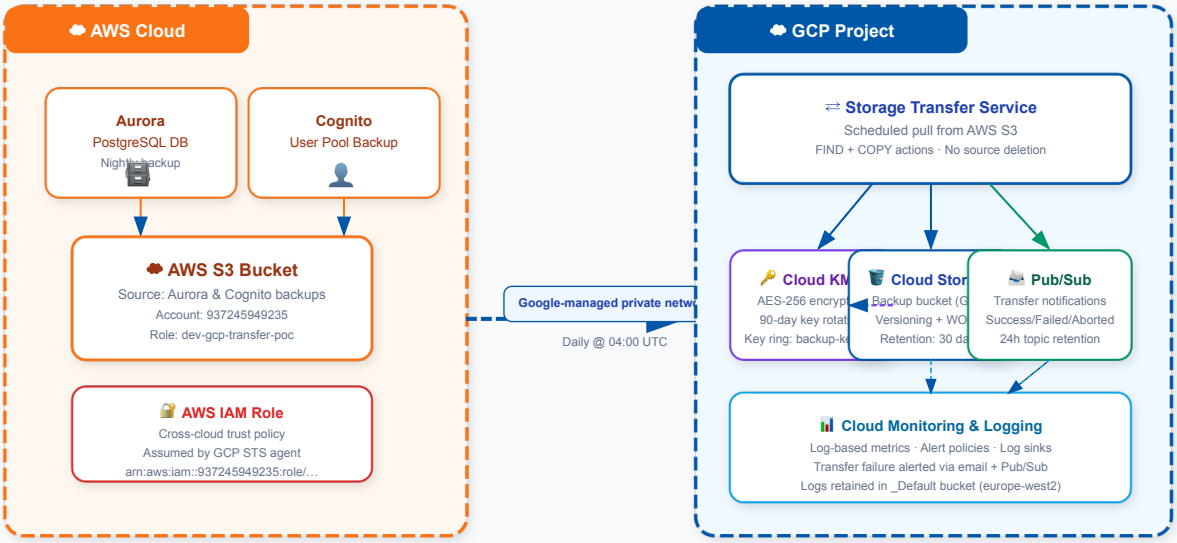
GCP Service	Purpose	Key Config
Cloud KMS	Dedicated encryption key for backup bucket	90-day auto-rotation, SYMMETRIC_ENCRYPTION
Cloud Storage	Immutable backup bucket with retention + WORM	STANDARD → NEARLINE at 30d, delete at 90d

Storage Transfer Service	Scheduled pull from AWS S3 via private network	Daily 04:00 UTC, no source deletion
Pub/Sub	Transfer status notifications (Success/Failed/Aborted)	24h retention, 7-day subscription
Cloud Logging	Transfer job audit logs & log-based metrics	FIND + COPY actions logged
Cloud Monitoring	Alerting on transfer failures	Email notification channel (optional)
IAM	Dedicated service account with least-privilege	storagetransfer.admin + logging.logWriter

2. High-Level Architecture Diagram

The diagram below shows the cross-cloud data flow from AWS to GCP, including the trust boundary between the two cloud providers via AWS IAM Role assumption.

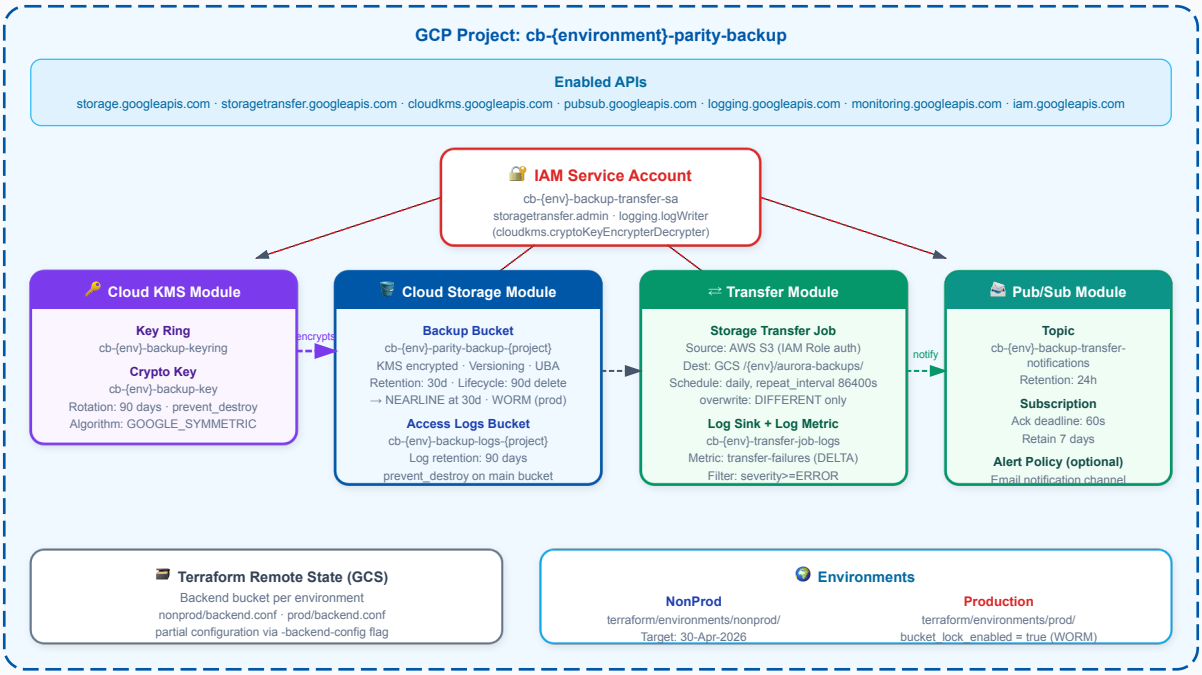
FIGURE 1 – CROSS-CLOUD ARCHITECTURE OVERVIEW



3. GCP Infrastructure Detail Diagram

This diagram shows all GCP resources deployed by Terraform, the IAM service account, and the relationships between modules.

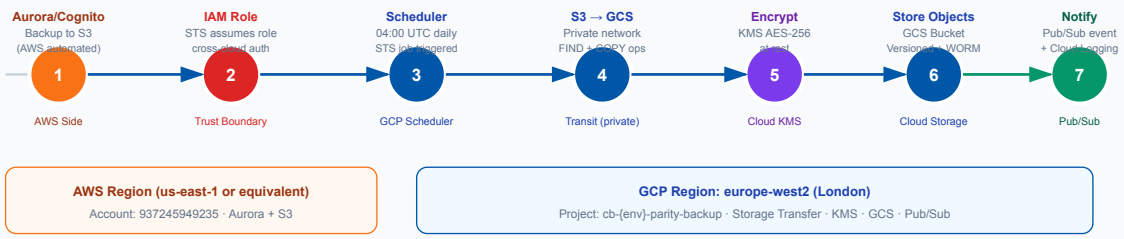
FIGURE 2 – GCP RESOURCE TOPOLOGY



4. Data Flow Diagram

The numbered steps below trace the end-to-end nightly backup transfer from Aurora PostgreSQL through to the encrypted GCS bucket, with monitoring events.

FIGURE 3 – NIGHTLY BACKUP TRANSFER DATA FLOW



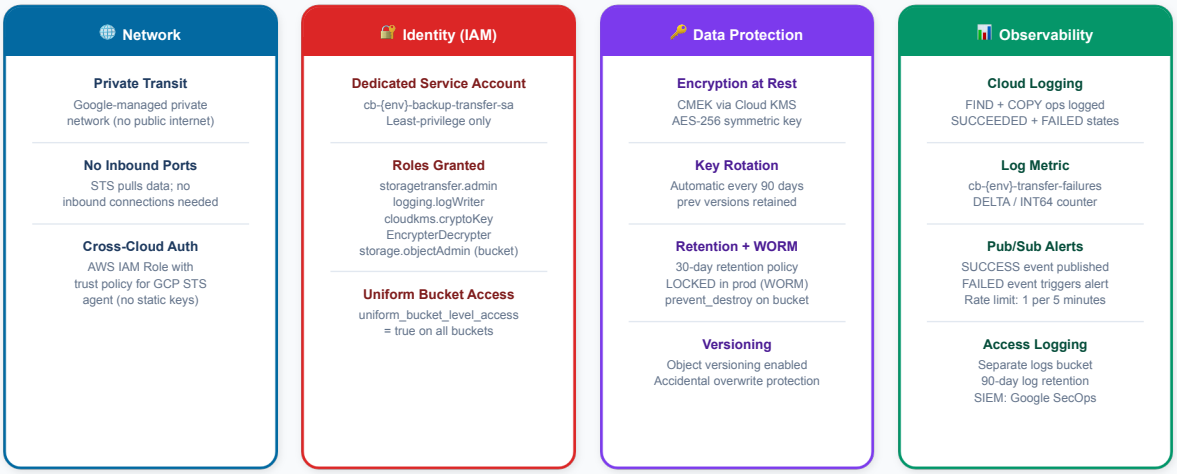
AWS Region (us-east-1 or equivalent)
Account: 937245949235 · Aurora + S3

GCP Region: europe-west2 (London)
Project: cb-{env}-parity-backup · Storage Transfer · KMS · GCS · Pub/Sub

5. Security Architecture

The solution implements defence-in-depth across network, identity, data, and operational layers.

FIGURE 4 – SECURITY CONTROLS OVERVIEW



Security Controls Summary

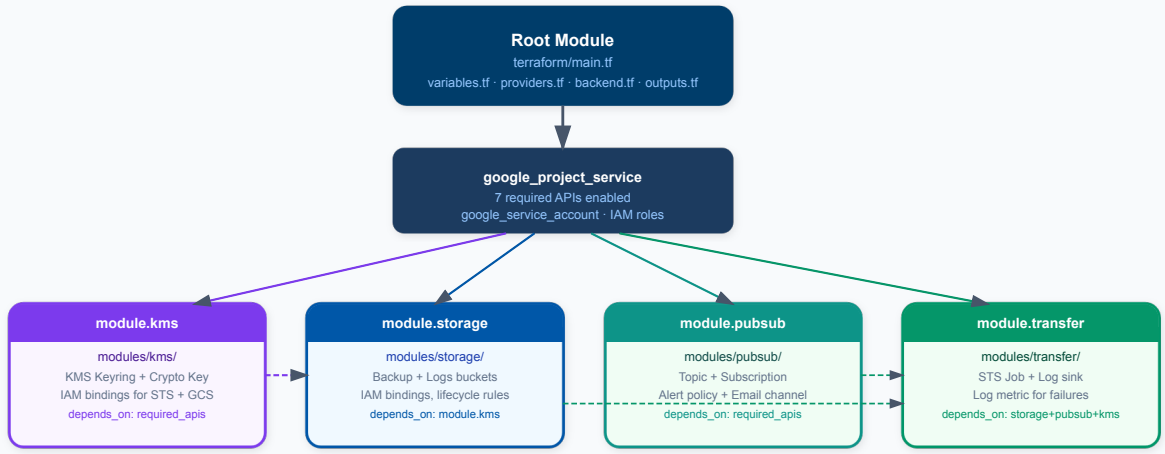
Control	Implementation	Status
Private network transfer	Google-managed private network (no public internet)	✅ Implemented
Encryption at rest	Cloud KMS CMEK (AES-256), 90-day rotation	✅ Implemented
Retention policy	30-day minimum retention on all backup objects	✅ Implemented
WORM / LOCKED mode	bucket_lock_enabled = true for production	⚠️ Prod only
Dedicated GCP project	Isolated project for parity backup storage	✅ Implemented

Uniform bucket-level access	uniform_bucket_level_access = true	✅ Implemented
Object versioning	versioning.enabled = true on backup bucket	✅ Implemented
Access logging	Separate bucket with 90-day log retention	✅ Implemented
SIEM monitoring	Google SecOps integration	🔵 Post-deploy
No static AWS keys	IAM Role assumption (STS federated auth)	✅ Implemented

6. Terraform Module Structure

The IaC is organised as a root module calling four child modules. Dependencies are explicitly declared via `depends_on` to ensure correct provisioning order.

FIGURE 5 – TERRAFORM MODULE DEPENDENCY GRAPH



Repository Structure

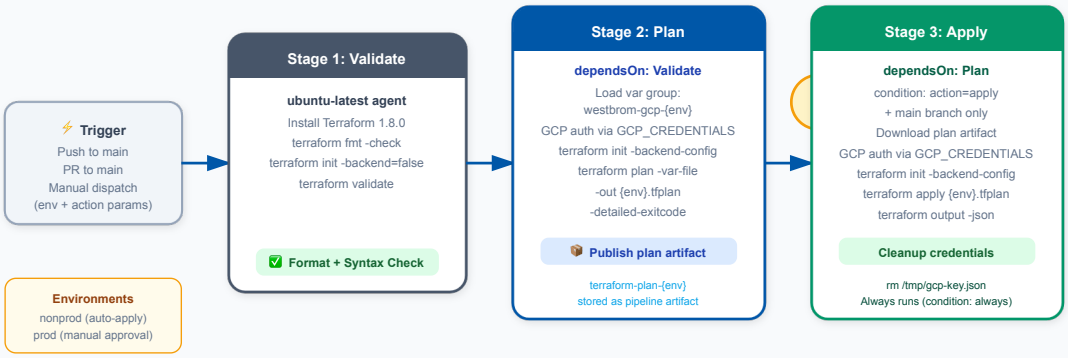
Path	Purpose
`terraform/main.tf`	Root module – wires all child modules, service account, APIs
`terraform/variables.tf`	All input variable declarations with defaults
`terraform/providers.tf`	Google provider configuration (v5+)
`terraform/backend.tf`	GCS remote state backend (partial config)
`terraform/outputs.tf`	Root outputs (bucket names, transfer job, etc.)
`terraform/modules/kms/`	Key ring, crypto key, IAM bindings
`terraform/modules/storage/`	Backup bucket + logs bucket, lifecycle rules, retention

`terraform/modules/pubsub/`	Notification topic, subscription, alert policy
`terraform/modules/transfer/`	STS job, log sink, log-based metric
`terraform/environments/nonprod/`	Non-prod tfvars + backend.conf
`terraform/environments/prod/`	Production tfvars + backend.conf (WORM enabled)
`scripts/`	init / plan / apply / validate shell helpers
`pipelines/azure-devops-pipeline.yml`	Azure DevOps YAML pipeline

7. CI/CD Pipeline Diagram

Deployments are managed via **Azure DevOps** with a three-stage pipeline. Production deployments require a manual approval gate.

FIGURE 6 – AZURE DEVOPS PIPELINE STAGES



Required Azure DevOps Variable Groups

Variable Group	Variable	Description
`westbrom-gcp-nonprod`	GCP_CREDENTIALS	Service account JSON for non-prod GCP project
`westbrom-gcp-prod`	GCP_CREDENTIALS	Service account JSON for production GCP project

8. Configuration Reference

Transfer Job Parameters

Parameter	Value	Notes
Source type	Amazon S3	Aurora PostgreSQL + Cognito backups
AWS IAM Role ARN (nonprod)	`arn:aws:iam::937245949235:role/dev-gcp-transfer-poc`	Cross-cloud role assumption
Destination	GCS – `{env}/aurora-backups/`	Encrypted with KMS CMEK
Schedule start	04:00 UTC	Configurable via variables
Repeat interval	86400s (24 hours)	Daily cadence
Network	Google-managed private	No public internet exposure
Overwrite policy	DIFFERENT only	Skip identical objects
Source deletion	false	Source objects preserved on AWS
Notification events	SUCCESS / FAILED / ABORTED	Published to Pub/Sub as JSON
Logging	FIND + COPY · SUCCEEDED + FAILED	Stored in Cloud Logging _Default

Storage Lifecycle Policy

Age	Action	Storage Class
0 – 30 days	Active storage	STANDARD

30 days	Transition (automatic)	NEARLINE
90 days	Delete (lifecycle rule)	—
Minimum retention	30 days (retention policy)	Cannot be deleted earlier

Key Assumptions & Decisions

Type	Description
ASSUMPTION	Backup retention: 30 days for nightly offsite backup
ASSUMPTION	Off-site backup only used in event of widespread AWS regional outage (RTO: 7 days, RPO: 24 hrs)
DECISION	Use GCP Storage Transfer Service with Google-managed private network
DECISION	Backup scope: Aurora DB Backup and Cognito Backup
DEPENDENCY	Deloitte GCP team will create the necessary GCP-side configurations
DEPENDENCY	Enable SIEM monitoring in Google SecOps post-deployment

Target Milestones

Milestone	Target Date	Status
Non-prod deployment	30 April 2026	⏳ In Progress
Production deployment	TBD	📋 Planned

9. Post-Deployment Checklist

Complete each item after a successful Terraform apply to validate the end-to-end solution.

Infrastructure Validation

- ☐ Verify GCP APIs are all enabled (7 services)
- ☐ Confirm KMS key ring and crypto key are created
- ☐ Verify backup bucket exists with correct naming
- ☐ Confirm backup bucket has retention policy (30 days)
- ☐ Verify versioning is enabled on backup bucket
- ☐ Confirm uniform bucket-level access is enabled
- ☐ Verify access logs bucket is created
- ☐ Confirm Pub/Sub topic and subscription are created
- ☐ Verify transfer job is created with correct schedule
- ☐ Confirm log sink and log metric are deployed

Functional Validation

- ☐ Test AWS IAM Role cross-cloud authentication
- ☐ Run first manual transfer to validate end-to-end flow
- ☐ Verify objects appear in GCS bucket with KMS encryption
- ☐ Check Cloud Logging for transfer operation logs
- ☐ Verify Pub/Sub notification on successful transfer
- ☐ Confirm objects are in correct path ({env}/aurora-backups/)
- ☐ Test monitoring alert by triggering a failed transfer
- ☐ Verify NEARLINE transition rule is configured
- ☐ Enable SIEM monitoring in Google SecOps
- ☐ Set `bucket_lock_enabled = true` for production WORM mode